

Chapter 3: Scatterplots and Correlation

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Part I: Exploring Data (The Basics)

Bivariate Data: Two Variables per Individual

- In Chapters 1 and 2, we examined biological data one variable at a time (e.g., *only* looking at a histogram of viral loads). We call this **univariate** data analysis.
- However, modern biological research rarely isolates a single measurement.
- A physiologist doesn't just care about a patient's blood pressure; they want to know if a patient's *blood pressure* changes in a predictable way based on their *sodium intake*.
- When we measure two different quantitative variables on the exact same individual, we are analyzing **bivariate** data.

Explanatory vs. Response Variables

Before we can mathematically map the relationship between two variables, we must determine their scientific roles:

- **Response Variable (y):** The outcome variable that measures the biological result of the study. It is the variable we are trying to predict or explain.
- **Explanatory Variable (x):** The variable that we suspect *explains* or *causes* changes in the response variable.

Note on Terminology

You will often hear researchers call x the *independent variable* or *features*, and y the *dependent variable* or *target variable*.

Examples 3.1 & 3.2: Assigning Variables

Ex 3.1: Enzyme Activity and Temperature

Students run an enzyme reaction at different water bath temperatures and measure how much oxygen gas is produced.

Variable	Measurement	Biological Role
Explanatory (x)	Temperature ($^{\circ}\text{C}$)	Students manipulate this to drive the reaction speed.
Response (y)	Oxygen Gas (mL)	The biological outcome being measured.

Ex 3.2: Forest Ecology and Plant Growth

An ecologist measures the percentage of open sky (canopy cover) and the height of the tallest fern in 50 forest plots.

Variable	Measurement	Biological Role
Explanatory (x)	Canopy Open Sky (%)	The physical environment is the underlying cause.
Response (y)	Fern Height (cm)	The biological growth response.

Examples 3.3 & 3.4: Context Matters!

Ex 3.3: Beer and Blood Alcohol

Student volunteers drank different amounts of beer, and a police officer measured their BAC.

- Explanatory (x): Number of beers consumed.
- Response (y): Percent alcohol in blood.

Ex 3.4: Height

The NCHS surveys Americans to document height, age, and sex. Here, there are no explanatory/response variables—just observation.

However, a pediatrician looking at the exact same data to assess a child's growth trajectory would use Age and Sex as **Explanatory** (x) variables and Height as the **Response** (y) variable.

Scatterplots

Once we have identified our explanatory (x) and response (y) variables, the most effective way to display their relationship is a **scatterplot**.

Rules for Building a Scatterplot:

- 1 Always plot the **Explanatory Variable on the horizontal (x) axis**.
- 2 Always plot the **Response Variable on the vertical (y) axis**.
- 3 Each individual appears as exactly **one dot** on the graph, located at the intersection of their x and y values.

Examining a Scatterplot

What to look for

In any graph of data, look for the overall pattern and for striking deviations from that pattern.

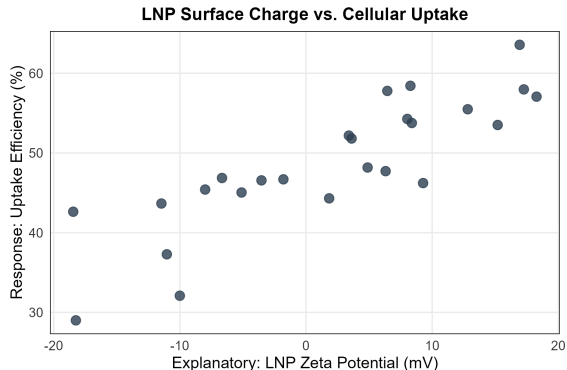
We describe the overall pattern of a scatterplot using four characteristics:

- **Form:** Is the relationship *Linear* (straight line) or *Non-Linear* (curved, U-shaped)?
- **Direction:** *Positive association* (slopes upward) or *Negative association* (slopes downward).
- **Strength:** How tightly do the points cluster around the underlying form? (*Strong, moderate, or weak*).
- **Outliers:** Are there points that fall far outside the overall cloud?

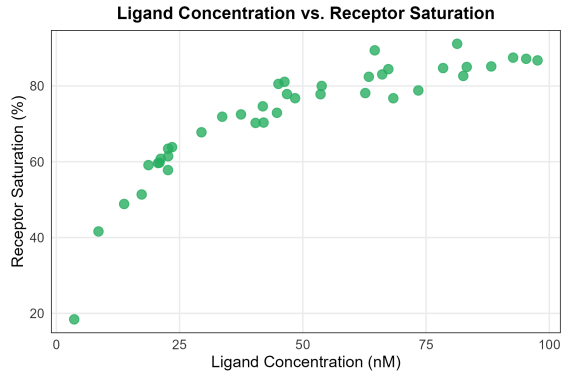
Example 3.5: Linear Relationships

Context: Pharmacologists testing if Lipid Nanoparticle (LNP) surface charge drives drug delivery efficiency.

- **Form:** Linear
- **Direction:** Positive
- **Strength:** Moderate-to-Strong
- **Outliers:** None apparent



Example 3.6: Non-Linear Biological Forms



Context: Measuring how a neurotransmitter drug binds to cellular receptors over time.

- **Form:** Strong, Non-Linear (asymptotic/hyperbolic).

Warning

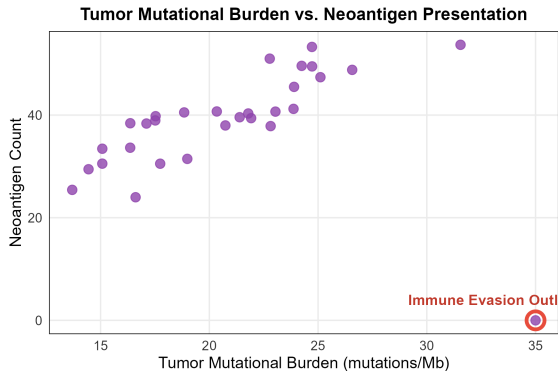
Standard linear statistical models cannot accurately summarize this data! The curve eventually flattens out near 100% because all receptors are full.

Example 3.7: Severe Bivariate Outliers

Context: High Tumor Mutational Burden (TMB) should logically trigger high Neoantigen presentation.

- **Overall Pattern:** Linear, positive, and moderately strong.
- **Outlier:** Notice the severe bivariate outlier at roughly (35, 0).

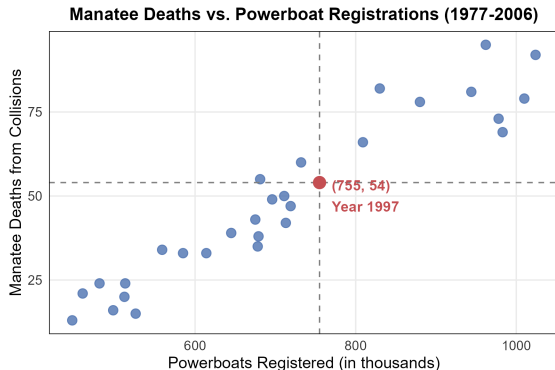
Why? Biologically, this specific tumor has mutated to delete its surface presentation machinery (HLA-loss), hiding from the immune system.



Positive vs. Negative Association

- Two variables are **positively associated** when above-average values of one variable tend to accompany above-average values of the other. (The cloud slopes UP).
- Two variables are **negatively associated** when above-average values of one variable tend to accompany below-average values of the other. (The cloud slopes DOWN).

Example 3.8: An Endangered Species (The Manatee)



Context: Does the number of registered powerboats (x) explain the number of manatee deaths (y) in Florida?

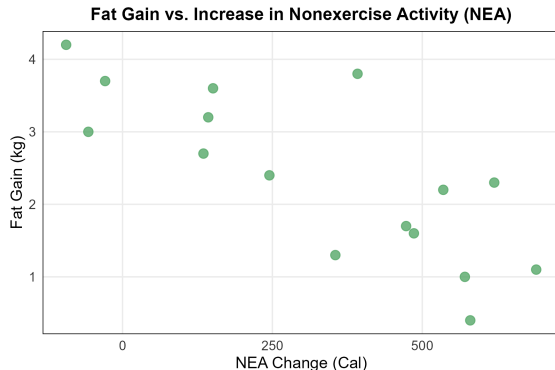
- **Direction:** Positive. Higher powerboat registrations accompany higher manatee deaths.
- **Form:** Linear.
- **Strength:** Strong.

Note: The plot is properly scaled, avoiding extra blank space on the edges.

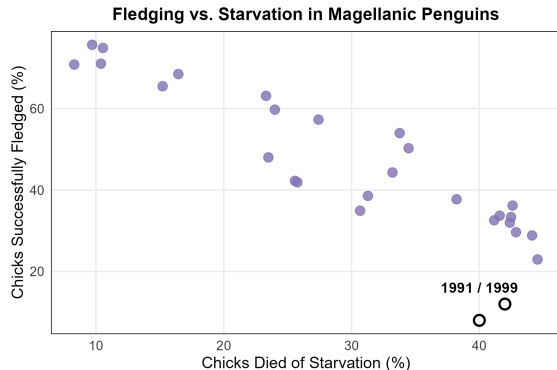
Example 3.9: Does Fidgeting Keep You Slim?

Context: Researchers overfed volunteers and measured their Nonexercise Activity (NEA) change (x) vs. Fat Gain (y).

- **Direction:** Negative. People with larger increases in NEA tend to gain *less* fat.
- **Form:** Linear.
- **Strength:** Moderately strong. The points make up a linear pattern but deviate quite a lot from the line.



Example 3.10: Magellanic Penguins



Context: Penguin chicks fledged (y) vs. chicks dying of starvation (x).

- **Trend:** Clear negative, linear pattern.
- **Outliers:** Two points (1991 and 1999) sit far below the pattern.

Why? Torrential, unusual rain killed over 40% of chicks in those specific years, breaking the normal starvation/fledging trend.

Correlation (r)

Scatterplots rely on subjective visual judgment. To make reproducible claims, we need a mathematical summary.

Pearson's Correlation Coefficient (r) measures the direction and exact strength of the *linear* relationship between two quantitative variables.

The formula standardizes both variables (converting them into z-scores), multiplies them together, and averages them:

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

Crucial Properties of r

- 1 r is always a number strictly between -1 and $+1$.
- 2 $r > 0$ indicates a positive linear association; $r < 0$ indicates a negative linear association.
- 3 Values near 0 indicate a very weak linear relationship. Values near -1 or $+1$ indicate points that lie nearly exactly on a straight line.
- 4 r **only measures linear strength**. It is completely blind to curved/non-linear relationships.

Example 3.11: The Mechanics of Correlation (Frog Locomotion)

To understand how the math works, let's look at a small dataset. A biologist measures the snout-vent length (cm) and maximum leap distance (cm) of 5 juvenile bullfrogs.

Frog ID	Length x (cm)	Leap Distance y (cm)
1	4.0	32.0
2	5.0	42.0
3	6.0	48.0
4	7.0	55.0
5	8.0	65.0

Example 3.11: Correlation in R (Methods 1 & 2)

Method 1: By Formula & Method 2: Custom Function

```
length_x <- c(4, 5, 6, 7, 8)
leap_y <- c(32, 42, 48, 55, 65)

# Method 1: Using mean() and sd() manually
n <- length(length_x)
z_x <- (length_x - mean(length_x)) / sd(length_x)
z_y <- (leap_y - mean(leap_y)) / sd(leap_y)
r_manual <- sum(z_x * z_y) / (n - 1)
r_manual
#> [1] 0.9931758

# Method 2: Wrapping into a custom function
calculate_r <- function(x, y) {
  n <- length(x)
  top <- (x - mean(x)) / sd(x)
  bottom <- (y - mean(y)) / sd(y)
  return(sum(top * bottom) / (n - 1))
}
calculate_r(length_x, leap_y)
#> [1] 0.9931758
```

Example 3.11: Correlation in R (Method 3)

While it's important to understand the math, R has a built-in shortcut for calculating Pearson's correlation coefficient.

Method 3: The fast way

```
# In practice, you will simply use the cor() function:  
cor(length_x, leap_y)
```

Output: [1] 0.9931758

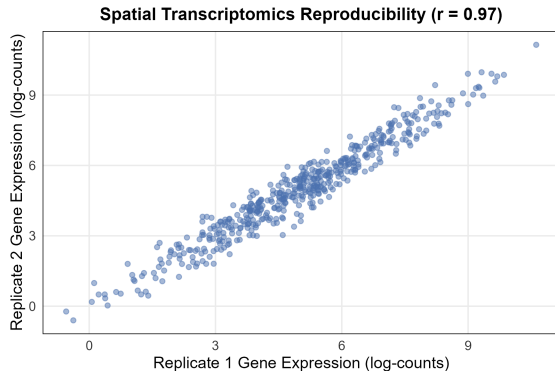
Interpretation: An r of roughly 0.993 tells us there is a nearly perfect, positive linear relationship between a juvenile frog's length and how far it can leap.

Example 3.12: Calculating r (Spatial Reproducibility)

Context: A lab runs a spatial transcriptomics chip twice. They calculate the correlation of gene expression levels (log-counts) across 500 genes between Replicate 1 and 2.

Interpretation:

An r of **0.9696** confirms a remarkably strong, positive linear relationship, proving the assay performs almost identically between independent runs.



Example 3.13: The Unit-Independence of r

Because the formula divides by standard deviations (s_x and s_y), it entirely strips away the physical units of measurement.

Rule

Correlation has no units, and changing the units of measurement does not change r .

If you calculate correlation between swimming speed (**km/h**) and energy expenditure (**Kcal**), you get $r = 0.9424$.

If a colleague converts the data to meters per second (**m/s**) and Joules (**J**), the correlation remains **identically 0.9424**.

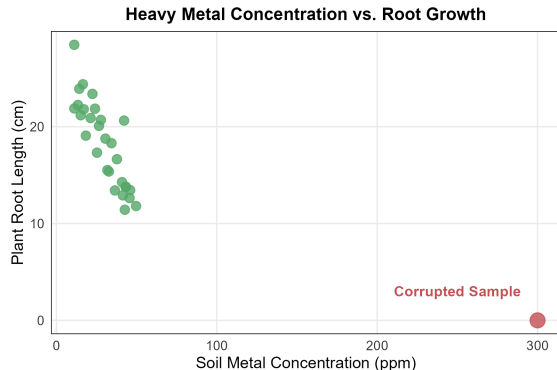
Example 3.14: The Non-Resistance of Correlation

Remember that the mean ($\bar{x}$) and standard deviation (s) are non-resistant to outliers.

Because the formula for r relies entirely on $\bar{x}$ and s , **correlation is strongly non-resistant to outliers.**

A single extreme point can completely destroy a strong correlation or artificially create one out of thin air! Let's look at a soil toxicology study...

Example 3.14: Heavy Metal vs. Root Length



29 normal samples show a moderate negative relationship ($r = -0.77$).

One single corrupted industrial sample was accidentally included.

By including just **one** anomalous data point out of 30, the correlation incorrectly jumps to **-0.95**, making it look nearly perfect!

The Golden Rule of Bivariate Data:

Never calculate a correlation coefficient without looking at the scatterplot first!

r cannot detect curves, and r is easily fooled by outliers.

Summary of R Commands (Chapter 3)

Here are the key functions introduced in this chapter for bivariate analysis:

R Command	What it Calculates
<code>length(x)</code>	Returns the number of elements (n) in a vector.
<code>sum(x * y)</code>	Multiplies two vectors together element-by-element, then sums the result.
<code>function(x, y) {...}</code>	Allows you to create your own custom, reusable mathematical formulas.
<code>cor(x, y)</code>	Calculates Pearson's Correlation Coefficient (r) between two variables instantly.